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UNDERGRADUATE INTERDISCIPLINARY RESEARCH WORK PROPOSAL:

THE DESIGN AND EVALUATION OF A DIGITAL NUDGE SYSTEM IN A BEHAVIOURAL ECONOMICS CONTEXT; IMPROVING ACADEMIC DISCIPLINE IN HIGHER LEARNING INSTITUTIONS

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INTRODUCTION

1.1 Background of study

Academic discipline is how consistently students attend lectures, submit assignments, meet deadlines, and manage their study time. It is one of the strongest predictors of success in university. Many higher education institutions, especially in developing countries like Ghana, continue to struggle with low levels of student engagement and rising academic procrastination. Research shows that students often fail to manage their academic responsibilities not because they lack ability, but because of behavioral challenges such as present bias, forgetfulness, overconfidence, and limited attention. (Rozental, Forsström, Hussoon, & Klingsieck, 2022) These predictable habits cause students to make short-term choices like scrolling through social media instead of studying even when they know it will harm their long-term goals. Behavioral economics provides a powerful explanation for these patterns and offers practical, low-cost solutions through "nudges," which gently guide people toward better decisions without forcing them.

Nudge theory, developed by Thaler Richard and Sunstein Cuss, argues that simple changes in the decision environment such as reminders, prompts, or clearer choices can significantly influence behavior (Thaler & Sunstein, 2008). In education, nudges have been shown to improve assignment submission rates, strengthen study routines, and even raise academic performance. For example, digital reminders have helped students overcome forgetfulness and pay more attention to academic tasks. Recent research also shows that digital nudges delivered through apps, learning management systems, or messaging platforms can reduce procrastination and keep students motivated, improving overall academic (Brandt & Oskorouchi, 2023).

With the rapid growth of educational technology in Africa, there is now an opportunity to apply behavioral insights directly through digital tools. Digital nudge systems make use of modern IT concepts such as user-friendly interfaces, push notifications, persuasive design, and data analytics to influence behavior in real time. Studies on digital behavior-support systems have shown that when technology is designed using behavioral principles, students respond more positively and are more likely to stay consistent in their academic habits (Mirsch, Lehrer, & Jung, 2017). Choice architecture tools like default settings, progress bars, personalized reminders, and goal-tracking features have been found to shape student behavior more effectively than traditional methods. In Ghana, where most university student own smartphones (Adzifome & Agyei, 2022), digital nudges offer a scalable and affordable way to support academic discipline.

However, while international research on digital nudging is growing, there is not enough evidence from Ghanaian institutions of higher learning such as the universities, where students face unique academic and behavioral pressures. Many Ghanaian students juggle school with work, experience high academic stress, or get easily distracted by digital platforms (Tetteh & Attiogbe, 2019). These challenges reduce their ability to consistently self-regulate. African studies highlight that digital tools informed by behavioral economics can significantly help in such environments, especially where academic support systems are limited. Still, the available evidence is scattered and inconsistent, making it necessary to carry out context-specific experimental studies to understand what actually works for Ghanaian students. (Mitchell & Jayasinghe, 2024)

University of Professional Studies, Accra (UPSA) is one of the top Universities in the Ghana where students enroll for higher studies. Due to its strategic nature in combining professional studies with regular studies it becomes ideal setting for studying a digital nudge system. UPSA is known for embracing technology in teaching and learning, and students frequently use LMS platforms, academic portals, and messaging systems. A digital nudge system designed for UPSA students providing timely reminders, motivational messages, and personalized cues can help address behavioral barriers that interfere with academic discipline. By applying behavioral economics principles such as salience, commitment, and social norms, such a system can guide students toward better academic decisions, especially during moments of distraction or indecision. An experimental evaluation is important for measuring whether these nudges truly improve academic discipline, and by how much.

As universities become more digital, there is a growing need for solutions that improve academic behavior without increasing administrative workload or costs. Digital nudges fit this need because they are easy to scale, inexpensive to maintain, and grounded in evidence. Research from UPSA can contribute significantly to behavioral science and digital education in Ghana by showing how students respond to different types of nudges such as deadline prompts, progress updates, or motivational cues. These findings can help universities design effective technology-based academic support systems that promote discipline and improve learning outcomes. Ultimately, studying a digital nudge system at UPSA not only fills a gap in local research but also demonstrates how behavioral economics and IT can work together to solve real academic challenges in an innovative and context-appropriate way.

1.2 Statement of the Problem

In many universities today, keeping students consistently disciplined by attending classes, submitting assignments on time, and staying engaged with course materials continues to be a major challenge (Essiam, 2019). At the University of Professional Studies, Accra (UPSA), informal observations and internal reports show that a number of students struggle with procrastination, missed deadlines, and low participation in digital learning platforms. These patterns do not only affect students' personal academic performance; they also make teaching and administration more stressful for lecturers. Traditional solutions like advising sessions, strict monitoring, or punitive deadlines often require a lot of time and resources, and they still do not fully address the behavioral reasons behind these academic challenges (Bijsmans & Schakel, 2018).

Behavioral economics helps explain why students sometimes fall behind even when they have the ability to do better. Biases such as present bias make students prioritize short-term pleasures like social media or relaxation over long-term academic goals. This bias, along with time-inconsistent behavior and the tendency to procrastinate, is common in student decision-making (Kleinberg & Oren, 2014). These challenges are not simply about laziness or lack of intelligence; they are predictable behavioral tendencies that can prevent students from following through on their intentions. Without targeted interventions, these behaviors continue and reduce students' ability to manage their studies effectively.

Nudge theory, introduced by Thaler and Sunstein (2008), offers a useful approach for addressing these issues. Instead of forcing students to behave in a certain way, nudges gently guide them by shaping how choices are presented. In education, digital nudges such as reminders, notifications about upcoming deadlines, motivational messages, or small prompts that show progress can be built directly into students' digital learning tools. Research shows that these nudges can help reduce procrastination, boost motivation, and improve academic discipline. For instance, digital reminders in online courses have been found to help students stay on track with assignments and study schedules (Garbers, Crinklaw, Brown, & Russell, 2023). Other studies show that personalized nudges increase engagement in large courses by helping students stay connected to their learning goals (Garbers, et al., 2025; Garbers, et al,, 2023). These findings suggest that well-designed nudges can play a meaningful role in improving students' academic behaviors.

However, the success of digital nudges is not guaranteed. Their impact can differ greatly depending on how they are designed, the timing of their delivery, and how students interact with digital platforms. (Koomar, 2024)This is especially important for Ghanaian universities like UPSA, where students' digital habits, device usage, and learning environments may look different from those in Western contexts where many nudge studies have been conducted.

Another key issue is that there is very little experimental evidence on the long-term impact of digital nudge systems in Ghanaian higher education. Very few studies have used rigorous methods like randomized controlled trials to compare the academic behaviors of students who receive digital nudges with those who do not. Without strong evidence, universities may not feel confident investing in digital nudging systems or may adopt systems that fail to produce real behavioral improvements.

Because of these gaps, it is important to examine the effectiveness of a digital nudge system within UPSA's own learning environment. This study will explore crucial questions: What types of nudges reminders, commitment prompts, or social comparison cues work best for UPSA students? To what extent do these nudges change academic behavior's, and do these effects last beyond the intervention period? Finally, what technological and ethical concerns arise when scaling a digital nudge system, including issues related to student privacy, notification fatigue, and system transparency?

By providing credible, locally grounded evidence, this research aims to support UPSA's efforts to improve student discipline while also contributing to the wider academic discussion on digital nudging in African higher education.

1.3 Study Objectives

1.3.1 General Objectives

To evaluate the effectiveness of a digital nudge app in improving academic discipline among UPSA students using behavioral economics principles and experimental evidence.

1.3.2 Specific Objectives

A. Theoretical objectives

I. To examine the effect of digital nudges on students' timely submission of assignments.

II. To assess the effect of digital nudges on students' consistency in class attendance.

B. Information Technology Objectives

III. To design and develop a functional digital nudge system that delivers behavioral prompts to students through an integrated platform.

IV. To pilot test the platform for usability, message timing, and data capture to ensure the intervention is technically feasible and privacy-aware before full RCT rollout.

1.4 Scope of the project

This study focuses on developing and evaluating a digital nudge application designed to enhance academic discipline among students at the University of Professional Studies, Accra (UPSA). The application will include features such as reminders, motivational prompts, and deadline alerts, and will also provide access to key LMS functions so that students can easily view course materials, submission deadlines, and academic tasks in one place. The app will be tested with a selected group of undergraduate students to assess how the nudges influence class attendance, assignment submission behavior, and overall academic engagement.

The study is limited to nudges delivered directly through the application, rather than through external channels like WhatsApp or email. It will not involve modifying UPSA's existing LMS infrastructure; instead, the app will simply integrate or link to LMS content for convenience. Data will be collected from in-app activity logs, assignment submission records, and feedback from participating students. Long-term behavioral changes and advanced AI-driven features are beyond the scope of this research.

Overall, the study is restricted to creating a functional prototype of the nudge application, testing its short-term impact on academic discipline, and analyzing student responses within the UPSA environment

1.5 Significance of the study

A. Theoretical Significance

This study adds to the fields of behavioral economics and educational technology by demonstrating how digital nudges can influence student behavior within a university setting. While nudging has been widely studied in areas like health, personal finance, and consumer decisions, there is still limited theoretical research on its role in promoting academic discipline particularly within African universities.

By creating and evaluating a digital nudge system that works alongside an LMS, this study provides a better understanding of how choice architecture functions when applied through digital tools students frequently use. It explores which forms of nudges such as reminders, notifications, motivational cues, or deadline alerts are most effective in shaping academic behaviors, thereby enriching nudge theory in educational contexts.

The results will help extend existing behavioral frameworks by showing how digital platforms, real-time feedback, and message framing can improve student engagement and self-management. Ultimately, this research deepens theoretical knowledge on how nudges operate in technology-driven learning environments and creates a basis for future studies on digital behavior-change systems in education.

B. Information Technology Significance

This study has important implications for information technology, particularly in the design and use of digital systems that support behavior change in education. By developing a custom digital nudge application integrated with the university's LMS, the study demonstrates how technology can be used not only to deliver content but also to shape user behavior in meaningful ways. This contributes to IT research by showing how system features such as automated reminders, push notifications, prompts, and embedded analytics can improve user engagement and adherence to academic tasks.

The project also highlights the practical value of lightweight behavioral systems that do not require complex artificial intelligence or large infrastructural changes. Instead, it illustrates how simple design elements timely notifications, intuitive dashboards, and user-friendly interfaces can significantly influence how students interact with academic platforms. These insights can guide IT professionals and educational institutions in building more responsive and behavior-aware systems.

Moreover, the study provides evidence on how integrating behavioral principles into digital platforms can enhance usability, improve interaction patterns, and support better decision-making among students. The findings can inform future development of digital learning tools, mobile applications, and LMS add-ons that embed behavioral design to boost academic discipline. Overall, the research contributes to the growing intersection of IT, human-computer interaction, and behavioral design by demonstrating the practical effectiveness of digital nudging in real educational environments.

1.6 Limitations of the project

This study may face several limitations. First, it will be conducted with a selected group of undergraduate students at UPSA, so the findings may not fully represent the behavior of the entire student population or students from other universities in Ghana. Second, the research focuses on the short-term impact of the digital nudge system, meaning it may not capture long-term behavioral changes or the sustainability of the observed effects. The effectiveness of the nudges also depends on students actively using the application and engaging with the prompts, and low participation or inconsistent use could affect the results. Additionally, some students may experience technical challenges, such as limited access to smartphones, internet connectivity issues, or unfamiliarity with digital tools, which may influence their interaction with the system. The results may also be context-specific, reflecting UPSA's academic environment, LMS structure, and student habits, which may limit generalizability. Finally, the study focuses on basic nudges such as reminders and motivational prompts and does not explore more advanced features like AI-based personalization or adaptive learning, which could potentially enhance the system's effectiveness

1.7 Organization of the study

This research is structured into five chapters, each designed to build a coherent and logical flow from the problem context to the final conclusions. **Chapter One** introduces the study by outlining the background, problem statement, research objectives, scope, significance, and limitations. **Chapter Two** reviews relevant literature on behavioral economics, nudge theory, digital academic technologies, and empirical evidence on digital nudges and academic discipline. **Chapter Three** presents the methodology, including the research design, sampling strategy, digital nudge system configuration, experimental procedures, data collection techniques, and analytical methods. It also explains the ethical considerations guiding the study. **Chapter Four** will provide the results and analysis of the randomized experiment, presenting findings on how different nudge types influence academic discipline. The chapter will interpret these results in relation to behavioral theory and previous studies. Finally, **Chapter Five** will summarize the major findings, discusses their implications, and offers recommendations for policymakers and university administrators. This chapter will also suggest areas for future research and concludes the study.

2. LITERATURE REVIEW

2.1 Introduction

The literature review examines existing studies and theoretical insights related to digital nudging and academic discipline. It highlights the principles of behavioral economics and nudge theory, examining how small, targeted interventions can influence students' decisions and promote behaviors such as submitting assignments on time, attending classes regularly, and engaging consistently with learning materials. The review also looks at research on digital nudge systems, focusing on their design, delivery methods, and effectiveness in educational settings. By analyzing these studies, the chapter identifies gaps in knowledge especially within African higher education and provides a foundation for applying a digital nudge application to enhance academic discipline at UPSA.

2.2 Theoretical Review

2.2.1 Behavioral Economics and Nudge Theory

The concept of digital nudging is grounded in behavioral economics**,** which highlights that individuals do not always make rational decisions. Their choices are shaped by cognitive biases, bounded rationality, and limited self-control. Thaler and Sunstein (2008) introduced nudge theory**,** arguing that designing the "choice architecture" of an environment can gently steer individuals toward better decisions without restricting freedom. In education, this means students do not necessarily lack the ability to engage in disciplined behaviors such as attending classes or submitting assignments on time but are often influenced by predictable behavioral tendencies like procrastination, present bias, forgetfulness, or a lack of salience for long-term academic goals.

2.2.2 Nudge Theory in Educational Contexts

Applying nudge theory to education involves shaping the digital choice architecture that students interact with. Koomar (2024) provides a practical framework for designing effective nudges in EdTech, emphasizing that the impact of nudges depends on context, message content, timing, and user segmentation. Not all interventions produce positive outcomes, highlighting the importance of careful design and evaluation (EdTech Hub Documentation).

Similarly, the study "Nudging in education: from theory towards guidelines for successful implementation" (Weijers, de Koning, & Paas, 2020)categorizes nudges into reminders, commitment devices, social norms, and feedback, linking each to cognitive processes such as memory, self-regulation, and social comparison. This guidance is crucial for designing a digital nudge app for UPSA students, ensuring interventions are targeted toward specific behavioral challenges like procrastination or attendance.

2.2.3 Empirical Evidence on Digital Nudging in Higher Education

Empirical studies further justify digital nudging in higher education. Garbers, Crinklaw, Brown, and Russell (2023) conducted a pilot randomized controlled trial in a graduate public health course, leveraging learning analytics to deliver behavioral nudges via an LMS. The nudges significantly increased student engagement, illustrating the theoretical link between data-driven interventions and behavioral economics principles.

Another study in TechTrends deployed tailored nudges based on students' intrinsic motivation profiles, demonstrating that nudges aligned with personal traits are more effective than generic reminders (Garbers, et al., 2025). These findings align with behavioral economic theory, emphasizing heterogeneous responses to different types of choice architectures.

2.2.4 IT and Human-Computer Interaction Foundations

Digital nudge systems also draw on persuasive technology and human-computer interaction (HCI) principles. Dimitrova and Mitrovic (2021) describe how interactive nudges such as visual prompts, reflections, and goal-setting scripts can be embedded in digital learning environments to guide student behavior. Open student models, which make learning data visible, act as a form of choice architecture that increases self-awareness, promotes reflection, and encourages social comparison. These IT-based mechanisms are directly relevant for designing a digital nudge app, which could use dashboards, progress bars, or notifications to subtly influence student decisions.

2.3 Review of Research Variables

This study investigates how a digital nudge system influences academic discipline among students at the University of Professional Studies, Accra (UPSA). The main variables: digital nudges, academic discipline, students' behavior, and user engagements are informed by behavioral economics and information systems literature.

2.3.1 Independent Variables

- Type of Nudge: This refers to the different behavioral prompts delivered by your digital nudge app (e.g., reminders, motivational messages, commitment prompts, social norm cues). Research shows that varying nudge types yield different behavioral effects for example, tailored nudges aligned with students' motivational profiles may influence engagement more strongly than generic reminders (Garbers, et al., 2025).

- Timing of Nudge Delivery**:** When the nudge is sent (e.g., before a deadline, weekly check-ins) can influence its effect. In a study of distance learners, nudges delivered shortly before deadlines significantly increased assignment submissions and class attendance.
- Nudge Frequency**:** How often nudges are sent (daily vs weekly) and how many are delivered during critical periods (e.g., assignment window) could moderate their effectiveness, as too many nudges may cause fatigue or annoyance. Persuasive systems literature suggests design must balance frequency and cognitive load.

2.3.2 Dependent Variables

- Student Engagement: This is a central outcome variable in many studies. Engagement can be quantified via LMS analytics (clicks, time spent, resource usage) as in the pilot RCT in public health education, where engagement tracked via learning analytics was used to assess the impact of behavioral nudges.
- Assignment Submission Rate: This measure whether nudges lead to more students submitting assignments and submitting them on time. For example, in the distance learning study, the number of students who submitted their assignments increased significantly after reminder nudges.
- Attendance**:** For courses with synchronous components (live sessions), attendance (or participation in chat) can be used. In the same study of working professionals, nudges before live classes raised attendance to full participation over time.

2.4 Contextual background of study area

The University of Professional Studies, Accra (UPSA) is one of Ghana's well-known public universities, widely recognized for its emphasis on business, professional, and applied social science education. In recent years, the school has invested heavily in digital learning tools, including a Learning Management System (LMS), online assessment platforms, and technology-supported communication channels. These upgrades have made teaching and learning more flexible, giving students easier access to course materials, assignments, and academic feedback.

However, even with these improvements, UPSA still faces several challenges related to students' academic discipline. Lecturers and administrators often observe issues such as late or missed assignment submissions, low activity on the LMS, irregular class attendance, and high levels of procrastination among undergraduates. These problems are usually not due to a lack of academic ability but are linked to behavioral factors like forgetfulness, low motivation, digital distractions, and difficulties in managing time or staying organized. Many students also balance school with work, commuting, or personal responsibilities, making it harder for them to keep up with academic demands (Tetteh & Attiogbe, 2019).

At the same time, UPSA's growing use of digital platforms presents an opportunity to test new strategies that support students' academic habits. Research from other countries has shown that digital nudges small, timely reminders or motivational prompts sent through apps or learning platforms can help students stay on track with studying, attending classes, and meeting deadlines. Yet, very little is known about whether these tools would work effectively in Ghanaian universities, since most existing studies have been done in foreign contexts.

This gap makes UPSA a suitable environment for studying how digital nudges might improve academic discipline. The student body is diverse, with varying levels of digital skills, motivation, and personal circumstances. UPSA's strong digital infrastructure also creates a practical setting to test a nudge-based application and see how it influences real student behavior. Findings from such a study could help the university enhance its academic support systems and contribute to broader discussions about digital learning innovations in Ghana.

In summary, UPSA provides an ideal context for examining the impact of a digital nudge system on academic discipline. This research connects behavioral economics, educational technology, and everyday academic challenges to offer valuable insights into improving student engagement and performance.

2.5 Review of Existing Systems and Technologies

Universities rely on a variety of digital systems to support teaching and learning, including Learning Management Systems (LMS), messaging platforms, cloud-storage tools, and digital calendars. At UPSA, these systems exist but are not cohesively deployed, resulting in a fragmented academic environment that places organizational burdens on students. The LMS provides a formal structure for academic content, but access challenges, outdated materials, and inconsistent usage limit its effectiveness. As a result, most course-related communication including slides, assignments, class reschedules, and announcements occurs through WhatsApp group chats managed by students and course representatives. Google Forms is frequently used for assessments, particularly when fee-based LMS restrictions prevent access to official platforms.

These systems illustrate a broader pattern in many African universities: digital tools exist, but they are not centralized or systematically managed. There is no unified calendar that automatically updates timetables or reflects rescheduled classes. Academic resources are distributed informally and manually, increasing the cognitive load on students who must constantly monitor multiple channels to remain academically organized. While tools such as Google Classroom, Canvas, and Microsoft Teams offer structured academic workflows, UPSA does not currently operate a single, integrated platform of this nature.

A critical limitation of the existing ecosystem is that it is distribution-based rather than system-based. That is, information flows through scattered channels rather than through an institutional system designed for structured academic management. This fragmentation undermines consistency, increases the likelihood of missed deadlines, and weakens students' academic discipline.

These shortcomings provide a compelling justification for a unified academic platform that centralizes all content, timetables, assessments, events, and reminders. Instead of integrating with the existing fragmented systems, the proposed digital nudge application replaces them by serving as the single authoritative academic environment, directly managed by lecturers, class representatives, and academic departments.

2.6 Proposed System

The proposed system is a unified digital academic platform designed to replace the fragmented channels through which students currently receive course materials, schedules, announcements, and assessment information. Rather than integrating with multiple tools such as WhatsApp groups, Google Forms links, and the university's LMS, the system functions as the single, centralized source for all academic activities. The platform consolidates timetables, class reschedules, lecture slides, assignments, academic events, and automated reminders, providing students with one reliable environment for managing their academic responsibilities. This unified design directly addresses the inefficiencies caused by fragmented communication channels and reduces the cognitive burden on students who must currently monitor several platforms simultaneously.

At its core, the system includes a structured academic resource repository managed by lecturers, class representatives, and departmental administrators. These stakeholders upload slides, assignment instructions, announcements, and rescheduling information directly into the platform, ensuring that academic content originates from a consistent and authoritative source. A built-in timetable and event management module allows departments to upload official schedules at the start of the semester, with any subsequent changes automatically reflected in all student calendars. This eliminates the need for students to manually construct or update their timetables and ensures that important schedule updates are never missed.

The platform also incorporates a dynamic digital nudge engine grounded in behavioral economics principles. This component provides timely reminders, motivational prompts, progress indicators, and commitment cues that are designed to counteract behavioral tendencies such as forgetfulness, procrastination, and limited attention, common challenges identified in the literature on academic discipline. By linking nudges directly to the student's academic calendar, upcoming deadlines, and recent activity patterns within the system, the nudge engine supports consistent academic engagement in a personalized and non-intrusive way.

The student-facing mobile application presents all of this information through an intuitive interface that prioritizes clarity, organization, and salience. The home dashboard summarizes daily and weekly academic activities, while the built-in calendar visualizes classes, deadlines, and university events. A centralized resource hub houses all slides and course materials, ensuring that students no longer require WhatsApp groups or external communication channels to access essential content. The design emphasizes minimal cognitive load by surfacing the most important academic tasks and providing subtle prompts that guide students toward timely action.

By centralizing academic content management and embedding evidence-based nudges within a single platform, the proposed system offers a comprehensive solution to the organizational and behavioral challenges faced by UPSA students. It provides the structural coherence missing from the current environment and creates an academic ecosystem that supports discipline, consistency, and improved learning outcomes. This approach not only enhances academic coordination but also creates the ideal conditions for experimentally evaluating the effects of digital nudges within a Ghanaian university context.

2.7 Chapter Summary

This chapter explored the technological tools that currently support digital nudging in education, paying particular attention to systems that help improve academic discipline and student behavior. It first reviewed major learning management systems (LMSs) like Moodle, Canvas, Sakai, and Blackboard, noting their features such as reminders, analytics, and communication tools that make digital nudging possible. The chapter also examined other digital nudge technologies, including Just-in-Time reminders, AI-based learning analytics, mobile push-notification apps, and behavioral messaging systems, showing how they apply ideas from behavioral economics and nudge theory. Findings from recent studies highlighted how elements like message framing, timing, frequency, personalization, and simple user interfaces can influence students' engagement, assignment submission habits, and self-management skills. The review further pointed out some gaps in existing technologies, such as limited adaptation to African university contexts, low levels of personalization, the possibility of nudge fatigue, and challenges integrating with institutional LMS platforms. Overall, the chapter provided a strong technological basis for developing the proposed digital nudge system for UPSA and emphasized the importance of context-specific, ethical, and data-driven digital tools for improving academic discipline.

3. METHODOLOGY

3.1 Introduction

This chapter outlines the methodology for evaluating the impact of a digital nudge system on academic discipline among UPSA students and for developing the proposed application. It begins with the research design, explaining the experimental approach that will be used to measure behavioral changes, followed by a description of the population, sample, and sampling techniques. The chapter also highlights the data collection instruments, including the digital nudge app, LMS logs, and questionnaires, and explains the system development methodology guiding the creation of the app. Furthermore, it addresses the crystallization of the problem, the requirements of the proposed system, and the overall system design. Together, these sections provide a clear and structured roadmap for conducting the study and implementing the digital nudge intervention.

3.2 Research Design

The study will adopt a causal, quantitative experimental research design to determine whether a digital nudge system can directly improve academic discipline among UPSA students. This design is suitable because the study aims to establish causation rather than simply describe behavior. Guided by the post-positivist paradigm, the research will assume that student behavior can be objectively measured and analyzed using LMS data and nudge-system records, allowing for valid causal conclusions.

A randomized controlled trial (RCT) will be used. Participating students will be randomly assigned to either a treatment group that receives behavioral nudges or a control group that uses the LMS without nudges. Random assignment will minimize bias and ensure both groups are comparable, making it possible to attribute any behavioral differences to the nudge intervention. This aligns with behavioral economics research, where experiments are commonly used to test the effects of choice architecture.

Although the study is primarily causal and quantitative, brief exploratory elements will be included. Short user-experience surveys will gather perceptions on ease of use, usefulness, and relevance of the nudges. These insights will help explain variations in nudge effectiveness and provide context for the quantitative results.

Behavioral indicators such as LMS logins, time spent on course content, assignment submission timeliness, and attendance will be captured automatically through system logs. These objective measures support rigorous statistical analysis, including mean comparisons and regression techniques. The design follows common approaches in persuasive technology and learning analytics, which stress controlled testing, measurable behavior, and user-centered intervention design.

3.3 Population

The target population for this study will consist of undergraduate students at the University of Professional Studies, Accra (UPSA). This population is appropriate because the research aims to evaluate the impact of a digital nudge system on academic discipline, including behaviors such as class attendance, timely assignment submission, and engagement with course materials areas directly relevant to undergraduate students who actively participate in the university's learning management system (LMS). Undergraduate students are ideal participants because they represent the largest cohort of active learners who face recurring academic deadlines and digital learning interactions, making them suitable for testing behavioral interventions through a digital platform. Additionally, focusing on UPSA undergraduates allows the study to generate context-specific insights into how nudges can influence academic behavior in a Ghanaian higher education setting, which is currently underrepresented in empirical research. This ensures that the findings will be both practically relevant and theoretically informative for applying nudge theory in similar educational contexts.

3.4 Sample and Sampling Techniques

The study will use a sample of 120 undergraduate students from UPSA, drawn from the department of economics and actuarial science and the department of information technology, ranging from 1st year to 2nd year students. This sample size is considered adequate for an experimental study because it allows for meaningful statistical comparisons between treatment and control groups while remaining manageable for monitoring engagement with the digital nudge system. Dividing the sample evenly, 60 students will be assigned to the treatment group receiving nudges through the application, and 60 students will form the control group, relying solely on the LMS. This size is informed by previous studies in educational nudging, which have shown that samples of 100–150 participants are sufficient to detect measurable behavioral changes while maintaining practical feasibility for data collection and system monitoring (Garbers, Crinklaw, Brown, & Russell, 2023)

A quota sampling technique will be used to ensure that participants are drawn proportionally from the key subgroups. This method is appropriate because it allows the researcher to intentionally balance the sample across these categories, reducing the likelihood of over- or under-representing any group that may differ in responsiveness to digital nudges. By setting quotas for the first- and second-year students from the IT and Economics and Actuarial Science departments, the study can better capture variations in academic behavior and technology use.

Since the study relies on primary data collected through the digital nudge application and LMS activity logs, there is no use of secondary or time-series data. However, the intervention will be conducted over one full academic semester to capture sufficient behavioral data, including assignment submissions, class attendance, and LMS engagement. This time frame is justified as it allows students to experience multiple academic deadlines and interactions with the nudge system, providing a realistic context for measuring short-term changes in academic discipline.

3.5 Data Collection Instruments

The study will employ multiple primary data collection instruments to capture both actual behavior and students' perceptions of the digital nudge system. The main source of data will be the digital nudge application, which will automatically track students' LMS interactions, including login frequency, time spent on learning materials, assignment submission times, and participation in online sessions. These system logs will provide accurate, real-time measures of student engagement and academic discipline, directly aligning with the study objectives.

Additionally, a structured questionnaire will be used to collect students' feedback on the usefulness, relevance, and motivational impact of the nudges. The questionnaire will be adapted from validated tools used in previous research on educational nudges and persuasive technology, such as those by Garbers, Crinklaw, Brown, and Russell (2023) and Plak et al. (2025). Adaptation will ensure that the questions are suitable for the UPSA context, the digital nudge app, and the specific academic behaviors being assessed.

To ensure validity, the questionnaire will be reviewed by academic supervisors and experts in behavioral economics and educational technology, confirming that the items accurately measure engagement, motivation, and nudge effectiveness. A pilot test with 10–15 students will also be conducted to refine question clarity, language, and response scales.

Reliability will be evaluated using **Cronbach's alpha** to assess the internal consistency of questionnaire items. The app and LMS system logs are inherently reliable as they objectively record student activity. By combining objective behavioral data with subjective survey responses, the study will obtain a comprehensive dataset to effectively assess the impact of the digital nudge system on academic discipline.

3.6 System Development Methodology

The system will be developed using the Agile Incremental Development Methodology, which is suitable for behavioral and educational technology systems that require iterative refinement based on user feedback. Agile supports continuous prototyping, frequent testing, and integration of behavioral insights that may evolve as the system is evaluated.

The methodology is structured into four iterative cycles: Planning, Design, Development, and Evaluation.

1. Planning Phase

This phase defines system requirements, identifies key stakeholders (students, lecturers, IT department), and outlines the core academic and nudge features. Storyboards and user stories are created to map expected student behaviors, nudge triggers, and use-case scenarios. Behavioral frameworks (salience, reminders, commitment devices, social norms) are translated into functional design elements and system rules.

2. Design Phase

This phase involves producing the system architecture, UI/UX layouts, nudge-flow diagrams, and integration structures. The design emphasizes simplicity, visibility of tasks, and cognitive load reduction. Behavioral design elements are embedded into the interface such as progress bars, color-coded deadlines, and default settings. Prototypes for the central dashboard, reminder system, and content aggregator are created and reviewed with test users.

3. Development Phase

Development proceeds in short cycles (sprints), each building an increment of the application. Early increments include the core academic-resource hub. Later increments introduce the nudge engine, notification system, analytics logging, and personalization rules. Back-end and front-end components are integrated progressively, ensuring that the system remains lightweight and mobile-friendly.

4. Evaluation Phase

Each increment undergoes both technical testing and behavioral pilot testing. Usability tests assess navigation, clarity, and cognitive load. Behavioral tests evaluate whether nudges are correctly triggered and whether students respond as expected. Feedback is incorporated into subsequent sprints. This iterative evaluation supports refinement of message timing, content framing, and notification frequency.

The Agile approach is appropriate because it allows the system to evolve through experimentation, user feedback, and behavioral evidence, producing a refined application suitable for UPSA’s academic environment.

3.7 Crystallization of the Problem

Digital nudging promises a low-cost, theory-driven way to improve student discipline, but the specific problem that this study will address is that we lack reliable, context-specific evidence showing which digital nudges work for Ghanaian university students and why. Many students struggle with predictable behavioral barriers procrastination, forgetfulness, and limited self-regulation that nudge theory (choice architecture) and behavioral economics directly target (Thaler & Sunstein, 2008),yet educational implementations of nudges must be carefully designed and evaluated because effects vary by message type, timing, and context (Weijers et al., 2020). Recent randomized trials and pilot studies show that analytics-informed nudges can increase engagement in higher education (Garbers et al., 2023) and that interactive, well-designed choice architectures (progress bars, reflective prompts) improve learning outcomes in digital environments (Dimitrova et al., 2021). At the same time, review evidence and practitioner briefs emphasize that implementations in low- and middle-income settings are fragmented and highly context-sensitive, calling for more locally grounded experiments (EdTech Hub, 2024). In short, there is a theoretical and technological basis for using a digital nudge app at UPSA, but a clear empirical gap remains: rigorous, context-specific experimental evidence is required to identify which nudge designs and delivery rules actually produce meaningful, scalable improvements in academic discipline among Ghanaian students.

3.8 Requirements of the Proposed System

System requirements are categorized into functional, non-functional, and behavioral requirements to align with both the technological and nudge-based objectives of the study.

A. Functional Requirements

1. Centralized Academic Dashboard
    - Display upcoming deadlines, classes, events, and tasks in a single view.
2. Content Aggregation Module
    - Provide access to lecture slides, assignment instructions, timetables, and important links.
3. Automated Reminder System
    - Send deadline alerts, class reminders, and periodic check-ins.
4. Digital Nudge Engine
    - Deliver motivational messages, social norm cues, commitment prompts, and salience-based nudges.
5. User Activity Logging
    - Track app usage, opened resources, viewed deadlines, and interaction patterns.
6. Personalization Rules
    - Adjust timing and type of nudges based on usage behavior (e.g., inactivity, missed reminders).
7. Progress Indicators
    - Display completion bars for weekly tasks, upcoming submissions, and academic goals.
8. Notification Manager

- Allow users to customize nudge categories and timing to reduce fatigue.

9. Link-Out Integration with LMS

    - Provide direct access to LMS pages, submission portals, and online assessments.

B. Non-Functional Requirements

1. Usability

    - Interface must be simple, intuitive, and accessible to students with varying digital literacy.

2. Performance

    - Application should load quickly, operate smoothly on low-end smartphones, and use minimal data.

3. Security & Privacy

    - Protect user data, comply with ethical standards, anonymize behavioral analytics.

4. Scalability

    - The system should support thousands of students without degradation in performance.

5. Reliability

    - Notifications and reminders must be delivered on time without failure.

6. Low Cognitive Load

- Interface design should reduce mental effort through clear layouts and salience cues.

C. Behavioral Requirements

1. Nudges must be non-intrusive and avoid notification fatigue.
2. Message framing must promote motivation, not guilt or pressure.
3. Nudges must be timely, especially before deadlines or class sessions.
4. Nudges must support autonomy, allowing opt-in/opt-out to preserve freedom of choice.

3.9 Design of the System

The system is designed using a layered architecture that separates behavioral logic, user interface, and academic content integration. The design embeds behavioral economics principles into both the system structure and the user interface.

A. System Architecture

The system consists of four primary layers:

- 1. Presentation Layer (Frontend Mobile App)
- Displays academic dashboard, calendars, reminders, and resources.
- Features a minimalist UI designed to emphasize salience and reduce cognitive load.
- Uses progress bars, color-coded deadlines, and structured layouts to guide student behavior.

2. Application Logic Layer

- Contains the Nudge Delivery Engine, which determines:
  - when nudges are triggered,
  - which type of nudge to deliver,
  - how nudges are framed.
- Manages the task scheduler, resource categorizer, and user settings.

3. Integration Layer

- Connects the system to external academic content (LMS links, Google Drive, portals).
- Provides structured APIs or secure link-outs to university academic tools.
- 4. Data Layer
- Stores:
  - user activity logs,
  - nudge delivery records,
  - academic events imported into the system,
  - personalization rules.
- Ensures data security, anonymization, and privacy compliance.

B. User Interface Design

The interface is structured around:

1. Home Dashboard
   - Displays next deadlines, next classes, and ongoing tasks.
   - Shows progress indicators for the week.
2. Resources Screen
   - Contains lecture slides, assignment details, timetables, and links.
3. Nudge Center
   - Shows delivered nudges and allows configuration of nudge types.
4. Calendar & Timeline View
   - Visualizes academic schedules, deadlines, and events.

The design uses visual nudges, such as highlighting urgent tasks and default sorting by importance.

C. Nudge Design and Logic

Nudges are categorized into:

- Reminders
- Motivational prompts
- Social norm nudges
- Commitment prompts

The Nudge Engine triggers these based on:

- Time until deadline,

- Inactivity on the app,
- Access to learning resources,
- Missed reminders,
- Weekly engagement patterns.

D. System Workflow

1. Student logs into the app.
2. System gathers academic data (deadlines, slides, announcements).
3. Dashboard displays tasks and schedules.
4. Nudge Engine analyzes behavior and schedules appropriate nudges.
5. Notifications are sent through push alerts.
6. User responses and activity are logged to improve personalization.

3.10 Chapter Summary

This chapter presented the methodological framework that will guide the study on how a digital nudge system affects academic discipline among UPSA students. It began by explaining the research design, which uses a causal, quantitative experimental approach to accurately assess the direct impact of digital nudges on student behavior. The chapter then outlined the target population, sampling techniques, and sample size, justifying the focus on undergraduate students and the use of random sampling to enhance representativeness and minimize bias

It also described the data collection instruments, including LMS system logs, activity records from the digital nudge application, and a structured questionnaire adapted from established behavioral-technology studies. Steps for ensuring the validity and reliability of these instruments such as expert evaluation, pilot testing, and the use of Cronbach's alpha were also discussed.

In addition, the chapter detailed the system development methodology, the problem definition, and the requirements and design of the digital nudge application, ensuring that the system is grounded in behavioral economics and educational technology concepts. Ethical considerations such as informed consent, data privacy, and participant protection were also addressed to uphold research integrity.

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